

1. Prerequisites

A knowledge of the specifications for P1, P2 and P3, their prerequisites and associated formulae, is assumed and may be tested.

2. Examination

- First assessment: June 2020.
- The assessment is 1 hour and 30 minutes.
- The assessment is out of 75 marks.
- Students must answer all questions.
- Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*.
- The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.

3. Notation and formulae

Students will be expected to understand the symbols outlined in *Appendix 7: Notation*.

Formulae that students are expected to know are given below and will **not** appear in the booklet *Mathematical Formulae and Statistical Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

This is a list of formulae that students are expected to remember and which will not be included in formulae booklets.

Vectors

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \bullet \begin{pmatrix} a \\ b \\ c \end{pmatrix} = xa + yb + zc$$

P4.3 Unit content

What students need to learn:		Guidance
1. Proof		
1.1	Proof by contradiction	Including proof of the irrationality of $\sqrt{2}$ and the infinity of primes, and application to unfamiliar proofs.